BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 205 · Tracing unfamiliar algorithms

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. A half adder receives 1 and 1. Give Sum and Carry.

Q2. What stays in a simplified expression after grouping four 1s?

Q3. How many items are in int[] a={2,3,5,8}?

Q4. Starting x=0, execute x+=3 three times. List x after each.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

trace table — Records variable values at successive execution steps.

accumulator — Variable storing a running total.

counter — Variable tracking how often an event occurs.

conditional — Instruction executed only when a Boolean test is satisfied.

iteration — Repeated execution of a statement block.

New learning

Choose trace columns — Track i, a[i] and total. Start with total=0.

Check the condition — Odd values 3 and 1 do not satisfy a[i] MOD 2 = 0; total stays 0.

Update then finish — 4 and 2 are even, so total becomes 4 then 6; the printed output is 6.

Worked example

Trace the sum of numbers greater than 3 in [2,5,4].

1. Predict and explain — Create columns index, value and total; begin at 0.

2. Check and extend — 2 is skipped, then 5 is added.

3. Check and extend — 4 is also added. Final total is 9.

We Do · Practise with guidance

Trace how many values are less than 5 in [6,2,4,7].

```
count=0;
for each value in [6,2,4,7]:
  if value < 5: count++
```

Predict and explain

Start count at zero. Value 6 is not below 5.

Check and extend

Values 2 and 4 are below 5, so count becomes 1 then 2.

7 is not below 5. Final result: 2.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

Trace total=0; for values [2,3,4], add each value. Give total after each.

Question 2

For [4,7,2,6], count values greater than 5. What is returned?

Question 3

An algorithm intends to count even numbers but uses if n%2==1. Identify the error and correct it.

Exit Ticket · Answer before checking

When tracing a loop, you can ignore the condition and update the accumulator every iteration.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. Sum=0, Carry=1.

Q2. Only variables that are constant across the group.

Q3. 4.

Q4. 3, 6, 9.

You Do pages · Independent questions

Q1. 2, 5, 9.

Q2. 2: the values 7 and 6 satisfy >5.

Q3. The test counts odd numbers. Use n%2==0 to count even numbers.

Exit Ticket · Model answer

False. Update only when the condition is true.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed: